

BACKLOG BUDDY ? STUDY MATERIAL

Signals and Systems ? Unit-wise Lecture Notes

Subject Code: EC301 Semester: 3

.....
Condensed notes covering all five units: Classification of Signals, LTI Systems and Convolution, Fourier Series and Transforms, Laplace and Z-Transforms, Sampling Theorem and Reconstruction.

Unit 1: Classification of Signals

- Definition, key concepts and significance of Classification of Signals
- Important diagrams / derivations / algorithms for Classification of Signals
- Solved examples and previous-year questions on Classification of Signals

Unit 2: LTI Systems and Convolution

- Definition, key concepts and significance of LTI Systems and Convolution
- Important diagrams / derivations / algorithms for LTI Systems and Convolution
- Solved examples and previous-year questions on LTI Systems and Convolution

Unit 3: Fourier Series and Transforms

- Definition, key concepts and significance of Fourier Series and Transforms
- Important diagrams / derivations / algorithms for Fourier Series and Transforms
- Solved examples and previous-year questions on Fourier Series and Transforms

Unit 4: Laplace and Z-Transforms

- Definition, key concepts and significance of Laplace and Z-Transforms
- Important diagrams / derivations / algorithms for Laplace and Z-Transforms
- Solved examples and previous-year questions on Laplace and Z-Transforms

Unit 5: Sampling Theorem and Reconstruction

- Definition, key concepts and significance of Sampling Theorem and Reconstruction
- Important diagrams / derivations / algorithms for Sampling Theorem and

Reconstruction

- Solved examples and previous-year questions on Sampling Theorem and Reconstruction

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory

problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.